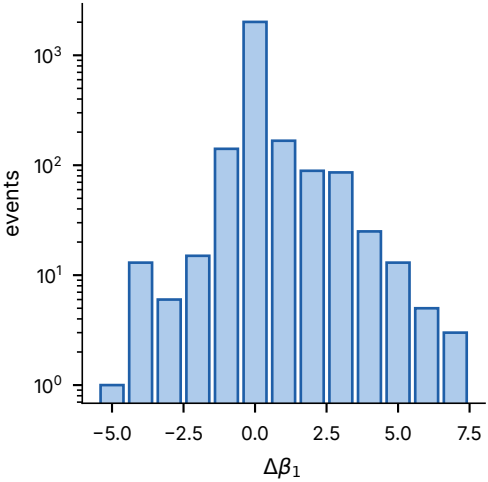
**a** $\Delta\beta_1$ over 2,576 $D$-events




**b** (dead edge?, new 2-cell?) → $\Delta\beta_1$

| dead | new 2c | up | same | down |
|---|---|---|---|---|
| no | no | 178 | 27 | 0 |
| no | yes | 210 | 1865 | 45 |
| yes | no | 0 | 2 | 0 |
| yes | yes | 0 | 118 | 131 |

(no, no) → down is empty: Theorem (cause)